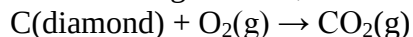


**CHAPTER 19**  
**Chemical Thermodynamic**

1. Which of the following represents an increase in entropy?  
A. Condensation of water  
C. crystallization of salt from a supersaturated solution.  
B. boiling of water.  
D. freezing of water.
2. Consider the following gas-phase reaction,  
$$2 \text{NO}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g}) + \text{O}_2(\text{g}) \quad \Delta G^\circ = +70.5 \text{ kJ}$$
  
If 1.00 mole of  $\text{NO}_2(\text{g})$  is placed in a 1.00-L flask (no  $\text{NO}(\text{g})$  or  $\text{O}_2(\text{g})$  initially) at  $25^\circ\text{C}$ , which of the following statements is **TRUE**?  
A. The reaction will occur spontaneously from left to right.  
B. The reaction will occur spontaneously from right to left.  
C. The reaction is not spontaneous in either direction.  
D. At equilibrium, the concentrations of  $\text{NO}(\text{g})$  and  $\text{O}_2(\text{g})$  will be much larger than the concentration of  $\text{NO}_2(\text{g})$
3. Calculate the change in entropy for the vaporization of ethanol ( $\Delta S_{\text{vap}}$ ) given that ethanol has  $\Delta H_{\text{vap}} = +45.59 \text{ kJ/mol}$  and a boiling point of  $79^\circ\text{C}$ .  
A. 130. J/K-mol      B. 577 J/K-mol      C. 0.58 J/K-mol      D. -577 J/K-mol
4. Calculate the standard entropy change for the following reaction,  
$$\text{Cu}(\text{s}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{CuO}(\text{s})$$
  
Given that  $S^\circ[\text{Cu}(\text{s})] = 33.15 \text{ J/K. mol}$ ,  $S^\circ[\text{O}_2(\text{g})] = 205.14 \text{ J/K. mol}$ , and  $S^\circ[\text{CuO}(\text{s})] = 42.63 \text{ J/K. mol}$   
A. 93.09 J/K      B. -195.66 J/K      C. -93.09 J/K      D. 195.66 J/K
5. Consider the following characteristics of chemical reactions,  
I. spontaneity  
II. maximum amount of work that can be done  
III. speed  
The value of  $\Delta G$  for a chemical reaction under a given set of conditions provides information about:  
A. I only      B. II only      C. I and II      D. I, II and III
6. For mercury the molar entropy of vaporization is  $92.92 \text{ J K}^{-1} \text{ mol}^{-1}$  and the molar enthalpy of vaporization is  $58.51 \text{ kJ mol}^{-1}$ . Therefore, the normal boiling point of mercury in  $^\circ\text{C}$  is:  
A. 357 $^\circ\text{C}$       B. 516  $^\circ\text{C}$       C. 670  $^\circ\text{C}$       D. 273  $^\circ\text{C}$
7. Which of the following statements best describes the spontaneity of a reaction at  $25^\circ\text{C}$  for which  $\Delta S_{\text{sys}}$  is negative and  $\Delta S_{\text{surr}}$  is positive?  
A. The reaction cannot be spontaneous at  $25^\circ\text{C}$ .  
B. The reaction will be spontaneous at all temperatures.  
C. The reaction will be spontaneous only if  $\Delta S_{\text{sys}}$  is greater in magnitude than  $\Delta S_{\text{surr}}$ .  
D. The reaction will be spontaneous only if  $\Delta S_{\text{surr}}$  is greater in magnitude than  $\Delta S_{\text{sys}}$ .
8. Which of the following do you expect to have the largest entropy at  $25^\circ\text{C}$ ?  
A. Fe(s)      B. CH<sub>4</sub>(g)      C. H<sub>2</sub>O(g)      D. Hg(L)
9. In which of the following reactions do you expect to have a decrease in entropy?  
A. Fe(s)  $\rightarrow$  Fe(l)      B. Fe(s) + S(s)  $\rightarrow$  FeS(s)  
C. 2 Fe(s) + 3/2 O<sub>2</sub>(g)  $\rightarrow$  Fe<sub>2</sub>O<sub>3</sub>(s)      D. HF(l)  $\rightarrow$  HF(g)
10. In which of the following reactions do you expect to have the smallest entropy change?  
A. 2 HF(g) + Cl<sub>2</sub>(g)  $\rightarrow$  2 HCl(g) + F<sub>2</sub>(g)      B. 2 Fe(s) + 3/2 O<sub>2</sub>(g)  $\rightarrow$  Fe<sub>2</sub>O<sub>3</sub>(s)  
C. CH<sub>4</sub>(g) + 2 O<sub>2</sub>(g)  $\rightarrow$  CO<sub>2</sub>(g) + 2 H<sub>2</sub>O(l)      D. Cu(s) + 1/2 O<sub>2</sub>(g)  $\rightarrow$  CuO(s)

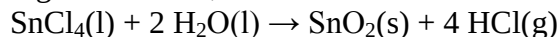
11. Calculate  $\Delta G^\circ$  for the following reaction,



Given that  $\Delta G_f^\circ[\text{CO}_2(\text{g})] = -394.4 \text{ kJ/mol}$  and  $\Delta G_f^\circ[\text{C(diamond)}] = +2.9 \text{ kJ/mol}$ .

- A. -397.3 kJ                      B. -394.4 kJ                      C. 397.3 kJ                      D. 394.4 kJ

12. Given the following information, calculate  $\Delta G^\circ_{\text{rxn}}$  for the following reaction at 25 °C.



Given that  $\Delta H^\circ_{\text{rxn}} = 133.0 \text{ kJ}$  and  $\Delta S^\circ_{\text{rxn}} = 401.5 \text{ J/K}$ .

- A. 252.6 kJ                      B. 122.9 kJ                      C. 13.4 kJ                      D. -252.6 kJ

13. The Third Law of Thermodynamics states that:

- A. all product-favored processes happen quickly.  
B. the total entropy in the universe is always increasing.  
C. the entropy of pure crystalline substances at 0 K = 0.  
D. the total energy in the universe is a constant.

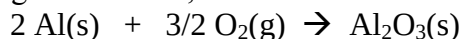
14. For the following general reaction,



Which of the following statements is correct?

- A. The reaction is spontaneous at all temperatures  
B. The reaction is spontaneous only at high temperature  
C. The reaction is non spontaneous at all temperature.  
D. The reaction is non spontaneous only at high temperature.

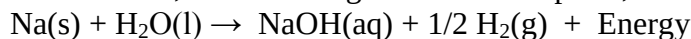
15. Given the following information, calculate  $\Delta G^\circ$  for the reaction below at 25 °C.



| Compound                           | $\Delta H^\circ$ (kJ/mol) | $\Delta S^\circ$ (J/mol K) |
|------------------------------------|---------------------------|----------------------------|
| Al(s)                              | ---                       | 28.3                       |
| O <sub>2</sub> (g)                 | ---                       | 205.1                      |
| Al <sub>2</sub> O <sub>3</sub> (s) | - 1676                    | 50.92                      |

- A. -1768 kJ                      B. -1668 kJ                      C. -1684 kJ                      D. -1583 kJ

16. When sodium is added to water, the following reaction takes place,



What can be said about this reaction?

- A. The reaction is not possible (non spontaneous at all temperature).  
B.  $\Delta G^\circ$  will increase with increasing temperature.  
C. The reaction will be non-spontaneous at low temperature.  
D.  $\Delta G^\circ$  will decrease with increasing temperature.

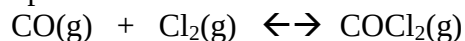
17. Consider the following reaction:



If the reaction is not spontaneous at room temperature, At what temperature will the reaction become spontaneous?

- A. 500 °C                      B. 0.5 K                      C. 501 K                      D. not at any temperature

18. The equilibrium constant  $K_p$  for the reaction



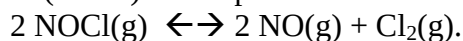
is  $5.62 \times 10^{35}$  at 25°C. Calculate  $\Delta G_f^\circ$  for  $\text{COCl}_2$  at 25°C. Given that  $\Delta G_f^\circ[\text{CO}(\text{g})] = -137.3 \text{ kJ/mol}$ .

- A. -341 kJ/mol                      B. 341 kJ/mol                      C. 682 kJ/mol                      D. 171 kJ/mol

19. For the reaction  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g})$ ,  $\Delta G^\circ = 173.4 \text{ kJ}$  at  $25^\circ\text{C}$ . The standard free energy of formation of NO is \_\_\_\_\_ kJ/mol.  
 A. 173.4                      B. 346.8                      C. 43.4                      D. 86.7
20. Isopropyl alcohol  $(\text{CH}_3)_2\text{CHOH}$  dissociates at higher temperatures.  
 $\text{CH}_3)_2\text{CHOH}(\text{g}) \rightarrow (\text{CH}_3)_2\text{CO}(\text{g}) + \text{H}_2(\text{g})$   
 At  $452.2 \text{ K}$  the equilibrium constant for the reaction is found to be 0.444. Calculate  $\Delta G^\circ$  for the reaction.  
A. 3.05 kJ                      B.  $3.05 \times 10^3 \text{ kJ}$                       C. 4.90 kJ                      D. -3.05 kJ
21. Consider the following reaction at  $298 \text{ K}$ :  
 $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{H}_2\text{O}(\text{l})$                        $\Delta H^\circ = -571.6 \text{ kJ/mol}$   
 Calculate  $\Delta S^\circ_{\text{univ}}$  for the reaction. (given that  $\Delta S^\circ$  for  $\text{H}_2\text{O}$ ,  $\text{O}_2$  and  $\text{H}_2$  are 69.9, 205 and 131 J/mol.K, respectively)  
 A.  $-327 \text{ J/mol.K}$                       B. 1591 J/mol.K                      C. 1918 J/mol.K                      D. zero
22. The following Lewis acid-base reaction has  $\Delta G^\circ = -39.9 \text{ kJ}$ .  
 $\text{Cd}^{2+}(\text{aq}) + 4 \text{NH}_3(\text{aq}) \rightarrow [\text{Cd}(\text{NH}_3)_4]^{2+}(\text{aq})$   
 What is the formation constant of  $[\text{Cd}(\text{NH}_3)_4]^{2+}$  at  $25^\circ\text{C}$ ?  
 A.  $-1.02 \times 10^7$                       B.  $1.02 \times 10^7$                       C.  $1.02 \times 10^{-7}$                       D.  $1.31 \times 10^{16}$
23. The second Law of Thermodynamics states that:  
 A. all product-favored processes happen quickly.  
B. the total entropy in the universe is always increasing.  
 C. the entropy of pure crystalline substances at  $0 \text{ K} = 0$ .  
 D. the total energy in the universe is a constant.
24. The equilibrium constant  $K_w$  for the auto-ionization of water,  $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$ , is  $1.0 \times 10^{-14}$ . What are the signs (+/-) of  $\Delta S^\circ$ , and  $\Delta G^\circ$  for the reaction at  $25^\circ\text{C}$ ?  
 A.  $\Delta S^\circ = (+)$ , and  $\Delta G^\circ = (-)$                       B.  $\Delta S^\circ = (+)$ , and  $\Delta G^\circ = (+)$   
 C.  $\Delta S^\circ = (-)$ , and  $\Delta G^\circ = (+)$                       D.  $\Delta S^\circ = (-)$ , and  $\Delta G^\circ = (-)$
25. Which one of following has a standard free energy equal to zero.  
 A.  $\text{CO}_2(\text{g})$                       B.  $\text{CH}_3\text{COCH}_3(\text{l})$                       C.  $\text{Cl}_2(\text{g})$                       D. ice

**Q26 and 27:**

Nitrosyl chloride (NOCl) decomposes at elevated temperatures according to the equation:



26. Calculate  $K_p$  for this reaction at  $227^\circ\text{C}$ . (For this reaction  $\Delta H^\circ = 81.2 \text{ kJ/mol}$ ,  $\Delta S^\circ = 128 \text{ J/K}\cdot\text{mol}$ )  
 A.  $4.90 \times 10^6$                       B. 62.8                      C.  $2.10 \times 10^{-7}$                       D.  $1.59 \times 10^{-2}$
27. If 1.0 atm of NOCl, 0.80 atm of NO, and 0.40 atm of  $\text{Cl}_2$  are mixed at  $227^\circ\text{C}$ , which of the following will occur?  
A. NO and  $\text{Cl}_2$  will be consumed (reactant-favored)                      B. no net reaction will occur  
 C. NOBr will decompose further (product-favored)                      D. can not be determined.
28. The second law of thermodynamics in mathematical form is \_\_\_\_\_.  
 A.  $\Delta S_{\text{sys}} + \Delta S_{\text{surr}} < 0$                       B.  $\Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$                       C.  $\Delta S_{\text{sys}} + \Delta S_{\text{surr}} \leq 0$                       D.  $\Delta S_{\text{univ}} \leq 0$

29. Arrange the following substances in the order of increasing entropy at 25°C.

|            |          |          |                      |                      |          |
|------------|----------|----------|----------------------|----------------------|----------|
| Substance  | HF(g)    | NaF(s)   | SiF <sub>4</sub> (g) | SiH <sub>4</sub> (g) | Al(s)    |
| Molar mass | 20 g/mol | 42 g/mol | 104 g/mol            | 32 g/mol             | 27 g/mol |

- A. SiF<sub>4</sub>(g) < NaF(s) < SiH<sub>4</sub>(g) < Al(s) < HF(g)  
B. HF(g) < Al(s) < NaF(s) < SiF<sub>4</sub>(g) < SiH<sub>4</sub>(g)  
C. Al(s) < NaF(s) < HF(g) < SiH<sub>4</sub>(g) < SiF<sub>4</sub>(g)  
D. Al(s) < HF(g) < NaF(s) < SiF<sub>4</sub>(g) < SiH<sub>4</sub>(g)